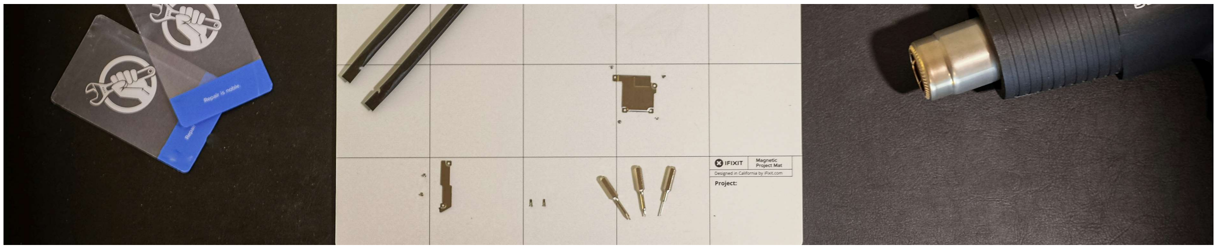

CA - Experiment 8 - Practical Exercise: Set Up a Jenkins CI Pipeline for a Maven Project, Use Ansible to Deploy Artifacts Generated by Jenkins.

This is a **detailed, step-by-step guide** to set up everything from **WSL installation** to **Jenkins Freestyle Project execution** with **Maven, Ansible, SSH setup (without password), and remote deployment**.

🎨 Step-by-Step Guide: Jenkins + Maven + Ansible CI/CD on Windows WSL 🚀

🔹 Overview

In this guide, we will: ✅ Install **WSL (Ubuntu on Windows)** 🐧
✅ Set up **Java, Maven, Git, and Ansible** 🛠️
✅ Configure **SSH connection between WSL & Remote Server** 🔑
✅ Set up **Jenkins (running on Windows) and integrate it with SSH & Ansible** 🔗
✅ Execute a **Jenkins Freestyle Project to deploy JAR files** 🎯

🟢 Step 1: Install WSL (Windows Subsystem for Linux)

📌 1.1 Enable WSL

1️⃣ **Open PowerShell as Administrator** and run:

```
wsl --install

```

2️⃣ Reboot your system if required.

📌 1.2 Install Ubuntu on WSL

1️⃣ Open **Microsoft Store** and install **Ubuntu 22.04 LTS**.
2️⃣ Launch **Ubuntu** and create a **username** and **password**.
3️⃣ **Update packages** inside Ubuntu:

```
sudo apt update && sudo apt upgrade -y

```

🟡 Step 2: Install Required Tools

📌 2.1 Install Java (JDK 17)

```
sudo apt install openjdk-17-jdk -y
java -version

```

✅ **Output should show** `openjdk version "17.0.x"`

📌 2.2 Install Maven

```
sudo apt install maven -y
mvn -version

```

✅ Ensure Maven is installed.

📌 2.3 Install Git

```
sudo apt install git -y
git --version

```

📌 2.4 Install Ansible

```
sudo apt update
sudo apt install ansible -y
ansible --version

```

🔵 Step 3: Set Up SSH Connection (No Password Required)

📌 3.1 Generate SSH Key

Run this inside **WSL (Ubuntu)**:

```
ssh-keygen -t rsa -b 4096

```

✅ **Press Enter** for default options.

> Why do we write **4096** ? **ssh-keygen -t rsa -b 4096**
>
> The `4096` in the command:
>
> ```
> ssh-keygen -t rsa -b 4096
>
> ```
>
> specifies the **key length** in bits.
>
> ---
>
> #### 🔹 Explanation of the Command:
>
> - `ssh-keygen` → The command to generate an SSH key pair.

- `-t rsa` → Specifies the **type** of key to generate (`RSA`).
- `-b 4096` → Specifies the **bit length** (i.e., key size).

🔹 Why Use 4096 Bits?

✅ **Stronger Security**:

- A **4096-bit key** is significantly more secure than the default **2048-bit key.**
- It provides better resistance against brute-force attacks.

✅ **Long-Term Security**:

- As computing power increases, **longer keys** ensure protection for a longer time.
- Many security professionals recommend **4096-bit RSA** for high-security environments.

✅ **SSH Best Practices**:

- While **2048-bit** is considered secure, **4096-bit** provides extra security without much performance loss.
- Modern systems can handle **4096-bit RSA** efficiently.

🔹 When Should You Use 2048 Instead of 4096?

- If performance is a concern (e.g., on embedded devices or old systems).
- If you need faster SSH authentication times.
- However, for **most servers and personal use**, **4096-bit is preferred**.

📌 **Conclusion:**
Using `4096` makes your SSH keys stronger and more future-proof, ensuring better security for your remote connections. 🚀🔒

📌 3.2 Copy Public Key to Remote Server

```
ssh-copy-id ronnie@172.27.158.79

```

✅ If prompted, enter the **remote user password**.

Finding the IP Address of the Remote Machine! 🛜💻🖥️👨‍💻

The IP address `172.27.158.79` is the **private IP address** of your remote machine (where Ansible is running). Let's break down how you obtained it:

🔹 How to Find the IP Address of the Remote Machine?

Since you are running **Windows WSL** and SSH-ing into a remote Linux machine (or another system), you likely found this IP address using one of the following methods:

🔸 Method 1: Using `ip a` or `ifconfig` on the Remote Linux Machine

1️⃣ **Log in to the remote machine** (where Ansible is deployed).
2️⃣ Run the following command:

```
ip a
```

OR

```
ifconfig
```

3️⃣ Look for an entry under `eth0` or `ensXXX` (depending on the network interface).

- Example output:

  ```
  2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
      inet 172.27.158.79/24 brd 172.27.158.255 scope global dynamic eth0
  ```

- Here, `inet 172.27.158.79` is the **IP address** of the machine.

🔸 Method 2: Using `hostname -I`

Run the following command on the remote machine:

```
hostname -I
```

This will return the machine's IP address, such as:

```
172.27.158.79
```

🔸 Method 3: Checking IP on WSL (for Windows Machines)

If you want to find your WSL (Windows Subsystem for Linux) instance's IP, run:

```
ip route | grep default
```

Example output:

```
default via 192.168.1.1 dev eth0 proto dhcp src 172.27.158.79 metric 100
```

Here, `172.27.158.79` is the **WSL machine's IP**.

🔹 How Was It Used in Jenkins?

- The IP address was used in **Jenkins SSH deployment**:

  ```
  scp "C:/Users/Saurav/.jenkins/workspace/Maven-Ansible-Deploy/target/*.jar" ronnie@172.27.158.79:/home/ronnie/deployment/
  ssh ronnie@172.27.158.79 "ansible-playbook /home/ronnie/ansible-deploy/deploy.yml"
  ```

- `ronnie@172.27.158.79` means:
  - `ronnie` → The **username** on the remote machine.

- `172.27.158.79` → The **IP address** of the remote machine.

🔹 Why Not Use `localhost` Instead?

- `localhost` refers to **your own system**, but your **Ansible machine** is on a separate system.
- If the remote machine is on the same network, you **must use its actual IP address** for SSH and SCP.

🔹 What If the IP Changes?

If your remote machine **restarts** or reconnects, the IP may change (if it's dynamic).
📌 To get a **static IP**, configure a **static IP address** in the network settings or use a **DNS name.**

✅ **Summary:**

- The **IP address** `172.27.158.79` was obtained by checking the remote machine's network configuration (`ip a` or `hostname -I`).
- It is used for **SSH connection from Jenkins** to copy files and run the Ansible playbook.
- If it changes, you can find the new IP using the same commands and update your Jenkins job accordingly. 🚀

📌 3.3 Test SSH Login

```
ssh ronnie@172.27.158.79

```

✅ If it logs in **without asking for a password**, SSH is correctly set up.

🟣 Step 4: Install and Configure Jenkins

📌 4.1 Install Jenkins on Windows

1️⃣ **Download Jenkins MSI** from [Jenkins Official Site].
2️⃣ Install Jenkins and start it as a **Windows Service**.
3️⃣ Open **Jenkins Web UI** at:

```
http://localhost:8080

```

📌 4.2 Unlock Jenkins

1️⃣ Copy the **initial admin password** from:

```
C:\Users\Saurav\.jenkins\secrets\initialAdminPassword

```

2️⃣ Paste it into the Jenkins setup page.

📌 4.3 Install Required Plugins

🔹 Go to **Manage Jenkins** → **Manage Plugins** and install:
✅ **Maven Integration Plugin**

✅ **SSH Agent Plugin**

🟠 Step 5: Create the Jenkins Freestyle Project

📌 5.1 Create a New Job

1️⃣ Go to **Jenkins Dashboard** → **New Item**
2️⃣ Select **Freestyle Project** → Name it **Maven-Ansible-Deploy**
3️⃣ Click **OK**

🟢 Step 6: Configure Jenkins Build Steps

📌 6.1 Set Up Source Code Management (Git)

1️⃣ Under **Source Code Management**, select **Git**
2️⃣ Enter your repository URL:

```
https://github.com/your-username/MVN-ANS-JEN-CICD.git

```

📌 6.2 Add Maven Build Step

1️⃣ **Go to Build** → **Add Build Step** → **Execute Windows Batch Command**
2️⃣ Paste this command:

```
cd %WORKSPACE%
mvn clean package

```

📌 6.3 Add SSH Deployment Step

1️⃣ **Go to Build** → **Add Build Step** → **Execute shell script on remote host using SSH**
2️⃣ Set **SSH site** as:

```
ronnie@172.27.158.79:22

```

3️⃣ **Enter the command to transfer and deploy JAR:**

```
scp "C:/Users/Saurav/.jenkins/workspace/Maven-Ansible-Deploy/target/*.jar" ronnie@172.27.158.79:/home/ronnie/deployment/
ssh ronnie@172.27.158.79 "ansible-playbook /home/ronnie/ansible-deploy/deploy.yml"

```

✅ **Save the job.**

> ℹ️ **Steps to add SSH Credentials in Jenkins.**
>
> ### 🔹 How to Add SSH in Jenkins for Remote Deployment
>
> To enable **Jenkins** to connect to a remote machine via **SSH**, follow these step-by-step instructions:
>
> ---

📌 Step 1: Install SSH Plugin in Jenkins

1️⃣ Open **Jenkins** (`http://localhost:8080`).
2️⃣ Go to **Manage Jenkins** → **Manage Plugins**.
3️⃣ Click on the **Available** tab and search for **"SSH Pipeline Steps"** or **"Publish Over SSH"**.
4️⃣ Select **Publish Over SSH** and click **Install Without Restart**.
5️⃣ Once installed, go back to the **Dashboard**.

📌 Step 2: Add SSH Key to Remote Server (Already done for remote machine ℹ️)

Since your Jenkins job runs SSH commands on the remote machine (`ronnie@172.27.158.79`), you need to add the **public key** to the remote server.

1️⃣ Copy the public key to the remote machine using:

```
ssh-copy-id ronnie@172.27.158.79
```

OR (if `ssh-copy-id` is not installed)

```
cat ~/.ssh/id_rsa.pub | ssh ronnie@172.27.158.79 "mkdir -p ~/.ssh && cat >> ~/.ssh/authorized_keys"
```

2️⃣ Verify SSH login without a password:

```
ssh ronnie@172.27.158.79
```

If successful, Jenkins can now SSH into the remote server without needing a password.

📌 Step 4: Configure SSH in Jenkins (Optional ℹ️)

1️⃣ Go to **Manage Jenkins** → **Manage Credentials**.
2️⃣ Under **Stores scoped to Jenkins**, click **(global)**.
3️⃣ Click **Add Credentials**.
4️⃣ Choose **Kind: SSH Username with Private Key**.
5️⃣ Set the following:

- **Username:** `ronnie`
- **Private Key:** Choose **Enter directly**, then paste the contents of:

  ```
  cat ~/.ssh/id_rsa
  ```

- **Passphrase:** Leave it empty. 6️⃣ Click **OK**.

📌 Step 5: Add SSH Host in Jenkins

1️⃣ Go to **Manage Jenkins** → **Configure System**.
2️⃣ Scroll down to **Publish Over SSH**.
3️⃣ Click **Add** under **SSH Servers**.
4️⃣ Fill in the details:

- **Name:** `RemoteServer`
- **Hostname:** `172.27.158.79`
- **Username:** `ronnie`
- **Remote Directory:** `/home/ronnie/deployment`
- **Use Password Authentication, or use a different key: Unchecked** 5️⃣ Click **Test Configuration** to verify the connection.

📌 Step 6: Configure Jenkins Job to Use SSH

1️⃣ Open your **Jenkins Freestyle Job** (`Maven-Ansible-Deploy`).
2️⃣ Click **Configure**.
3️⃣ *Under* **Build Steps**, *add:*

- **Execute shell script on remote host using SSH**.
  4️⃣ Enter the SSH command to copy and deploy the JAR file:

```
scp "C:/Users/Saurav/.jenkins/workspace/Maven-Ansible-Deploy/target/*.jar"
ronnie@172.27.158.79:/home/ronnie/deployment/
ssh ronnie@172.27.158.79 "ansible-playbook /home/ronnie/ansible-deploy/deploy.yml"

```

5️⃣ Click **Save**.

📌 Step 7: Test the Setup

1️⃣ Run the **Jenkins job**.
2️⃣ Check the **Console Output** for SSH connection success.
3️⃣ If everything works, Jenkins should:

- Copy the `.jar` file via SCP.
- Trigger the Ansible playbook to deploy the application.

✅ **Done!** 🎉 Your Jenkins pipeline now connects to the remote server over SSH for deployment. 🚀

🔵 Step 7: Create Ansible Playbook for Deployment

📌 7.1 Create Deployment Directory

On the **remote server**, run:

```
mkdir -p /home/ronnie/deployment

```

The `-p` option in the `mkdir` command stands for **"parents"**, and it ensures that:

✅ If the parent directories in the given path **do not exist**, they will be created automatically.
✅ If the directory **already exists**, no error will be thrown.

Example:

```
mkdir -p /home/ronnie/deployment

```

- 🔹 If `/home/ronnie` already exists, only `deployment` will be created.
- 🔹 If neither `/home/ronnie` nor `deployment` exists, both will be created.
- 🔹 If `deployment` already exists, the command does **nothing** (no error).

This is useful to **avoid errors** when creating directories in scripts or automation! 🚀

📌 7.2 Write Ansible Playbook

1️⃣ **Create** the playbook:

```
nano /home/ronnie/ansible-deploy/deploy.yml

```

2️⃣ **Paste the following YAML script**:

```yaml
---
- hosts: localhost
  tasks:
    - name: Ensure deployment directory exists
      file:
        path: /home/ronnie/deployment
        state: directory
        mode: '0755'

    - name: Copy the JAR file to the deployment folder
      copy:
        src: /home/ronnie/deployment/*.jar
        dest: /home/ronnie/deployment/

```


3️⃣ **Save and Exit** (`CTRL + X`, then `Y`, then `Enter`).

📌 7.3 Test Ansible Playbook

```
ansible-playbook /home/ronnie/ansible-deploy/deploy.yml

```

🟣 Step 8: Run the Jenkins Build 🚀

1️⃣ **Go to Jenkins Dashboard → Maven-Ansible-Deploy**
2️⃣ Click **Build Now** ▶️
3️⃣ **Jenkins will:** ✅ Clone your GitHub repo
✅ Build the **Maven project**
✅ Deploy the **JAR file using SSH & Ansible**

4️⃣ **Check remote deployment:**

```
ls -l /home/ronnie/deployment

```

✅ You should see the **JAR file** inside `/home/ronnie/deployment/`!
